ψ-Field Navigation, Positioning, and Timing Systems

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Cross-Reference to Related Applications

Pursuant to 35 U.S.C. §119(e), this application claims the benefit of the following U.S. provisional applications, the entirety of each being incorporated herein by reference:

- U.S. Provisional App. No. **63/865,279** (Confirmation No. 5785), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Gravitational and Signal Correction.*
- U.S. Provisional App. No. **63/865,283** (Confirmation No. 5905), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Energy Transmission.*
- U.S. Provisional App. No. **63/865,293** (Confirmation No. 6649), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Information Processing, Communication, and Advanced Computing.*
- U.S. Provisional App. No. **63/868,073** (Confirmation No. 8739), filed **Aug. 21, 2025**, entitled *Methods, Systems, and Apparatus for Artificial Generation, Control, and Application of Scalar Density Fields.*
- U.S. Provisional App. No. **63/893,633** (Confirmation No. 5179), filed **Oct. 4, 2025**, entitled *Systems and Methods for Electromagnetic Coupling, Actuation, and Application of Scalar Refractive Fields.*

This application also incorporates by reference, in their entirety for all purposes, **all other provisional and non-provisional applications filed by the same inventor on or after August 15, 2025**, including any continuations, divisionals, or continuations-in-part thereof. The foregoing applications are incorporated by reference in their entirety for all purposes, including definitions, figures, and embodiments. To the extent of any inconsistency, the present disclosure controls.

Statement Regarding Federally Sponsored Research or Development

No federal funding was used in connection with the inventions described herein. No government rights apply.

Incorporation by Reference (Additional Materials)

In addition to the applications identified above, technical materials authored by the inventor concerning scalar refractive field (ψ) dynamics, cavity–atom LPI tests, navigation/timing corrections, energy coupling, communication architectures, and quantum control are incorporated

by reference. All essential material for the claimed inventions is included explicitly herein; incorporated materials are provided for background and alternative implementations.

Definitions; Conventions

As used herein, "comprising" is open-ended and does not exclude additional elements or steps; "including" and "having" are used in a like manner. Numerical ranges include all sub-ranges and individual values unless otherwise specified. Singular terms include the plural and vice versa unless context dictates otherwise. Functional descriptions are supported by the corresponding structures, algorithms, and examples disclosed herein; no limitation to "means-plus-function" under 35 U.S.C. §112(f) is intended unless expressly recited. Throughout, the scalar refractive field is denoted ψ, with optical index $n = e^{\psi}$ where applicable.

No Admission

Any discussion of references, techniques, or prior work in the Background is provided solely for context and is not an admission that any such material constitutes prior art, is known, conventional, or required.

Abstract

Disclosed are systems and methods for navigation, positioning, and precision timing using controlled scalar refractive fields (ψ). Position and time are derived from the geometric and phase structure of a ψ corridor, rather than from remote clock synchronization or external electromagnetic signals. The approach yields deterministic propagation delays, synchronization-free timing, and geometry-locked coordinate recovery immune to spoofing or multipath.

Field of the Invention

This invention relates to navigation and timekeeping. Specifically, it enables autonomous determination of position, velocity, and time through measurement of a scalar refractive field (ψ) geometry, suitable for terrestrial, space, or underwater domains.

Background

Conventional Global Navigation Satellite Systems (GNSS) depend on remote clock synchronization, line-of-sight propagation, and external calibration. Such systems degrade under jamming, spoofing, or occlusion. A geometry-bound refractive corridor defined by ψ variations produces an intrinsic one-way delay $\tau = \frac{1}{c}\int e^{\psi}\mathrm{d}s$ that encodes spatial displacement. By measuring local ψ and its gradients, receivers can recover absolute position and time offset without relying on two-way exchanges or external clocks.

Summary

A ψ-based navigation network comprises a constellation of refractive field emitters generating overlapping corridors. Each corridor defines a unique path-dependent delay surface. A receiver measures the local ψ potential and gradient from at least three independent emitters to determine coordinates via inversion of intersecting delay surfaces. Timing is derived from the phase

of the ψ oscillation relative to a reference manifold. Because each corridor is geometrically distinct and non-replicable, the system provides native anti-spoofing and high-precision ranging without clock alignment.

Representative Claims

1. A navigation system comprising at least three scalar refractive field emitters defining overlapping ψ corridors, each having a known geometric signature, and a receiver configured to measure local ψ and $\nabla\psi$ to compute position and time.
2. The system of claim 1 wherein timing is recovered from the measured phase of ψ oscillations at the receiver relative to a reference manifold phase pattern.
3. The system of claim 1 wherein position is computed from the intersection of iso-ψ surfaces corresponding to known emitter geometries.
4. The system of claim 1 wherein authentication is achieved by verifying the local ψ curvature tensor against a stored template.
5. A method for synchronization-free ranging by integrating the refractive delay $\tau = \frac{1}{c}\int e^{\psi}\mathrm{d}s$ along a measured ψ corridor.
6. A hybrid navigation and timing system combining electromagnetic GNSS signals with ψ-based geometric correction to remove multipath and timing bias.

Brief Description of the Drawings

Fig. 1 – ψ-based navigation concept.
Fig. 2 – Iso-ψ surfaces and position recovery.
Fig. 3 – Phase-based timing extraction.
Fig. 4 – Spoof rejection and hybrid integration.

Detailed Description

1. Concept Overview

Each emitter generates a spatial ψ distribution $\psi_i(\mathbf{r})$ forming a refractive corridor. Receivers triangulate from multiple ψ_i intersections.

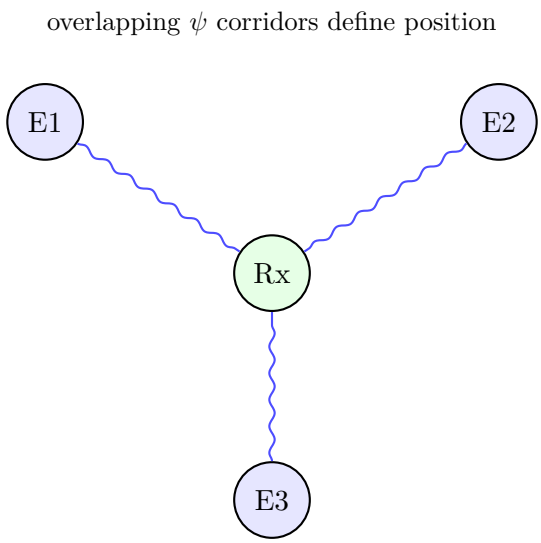

Figure 1: Three emitters form intersecting ψ corridors; the receiver position is determined by the intersection of the delay surfaces.

2. Iso-ψ Surfaces

Each emitter defines concentric iso-ψ surfaces. Position corresponds to intersection of at least three distinct ψ_i levels.

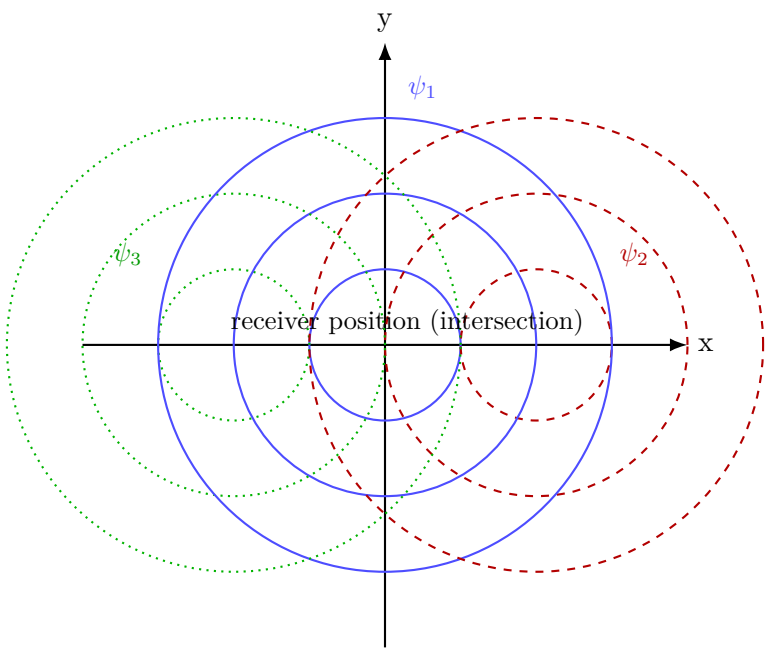

Figure 2: Iso-ψ surfaces from three emitters intersect at the receiver location, defining position geometrically.

3. Phase-Based Timing Extraction

The ψ oscillation phase $\phi(t) = \omega t + \psi(\mathbf{r})$ encodes absolute time. Local phase measurement yields clock-free timing.

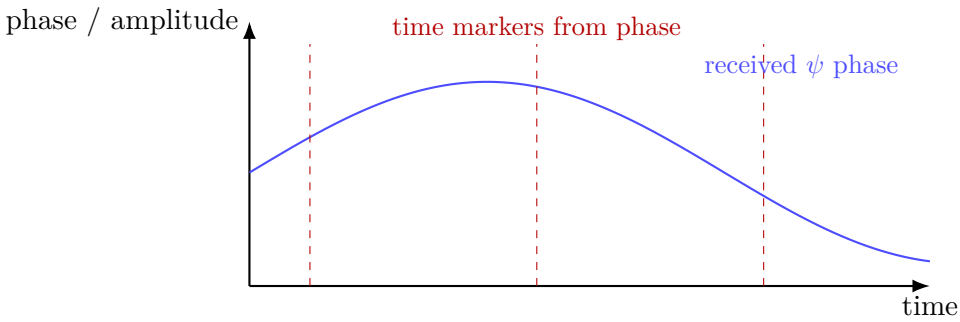

Figure 3: Phase-based timing extraction: local ψ oscillations provide clock-free absolute timing.

4. Spoof Rejection and Hybrid Integration

Because ψ geometry is unique, spoofed signals cannot replicate the measured curvature tensor $K_{ij} = \partial_i \partial_j \psi$. Hybrid integration uses GNSS as coarse input with ψ corrections.

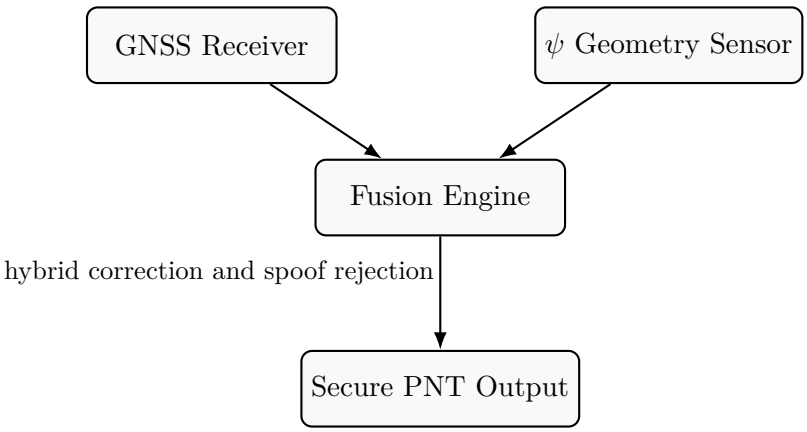

Figure 4: Hybrid navigation: ψ geometry corrects GNSS bias and rejects spoofing via curvature comparison.

Alternative Embodiments

(i) Terrestrial towers emitting stabilized ψ corridors for urban navigation under GNSS denial.
(ii) Deep-space arrays defining large-scale ψ manifolds for autonomous spacecraft navigation.
(iii) Submarine navigation using density-coupled ψ propagation through water.
(iv) Inertial fusion: blending ψ gradients with inertial sensors for drift-free tracking.
(v) Timing arrays embedded in power grids for synchronized industrial operations.

Industrial Applicability

Applicable to autonomous vehicles, aerospace navigation, undersea mapping, precision agriculture, military timing, telecommunications synchronization, and GNSS-independent PNT infrastructures.

Conclusion

ψ-based navigation and timing systems provide absolute position and time derived directly from field geometry, eliminating dependence on synchronized clocks or vulnerable EM signals. The approach yields inherent spoof resistance, deterministic delay, and geometric self-verification, establishing a new paradigm for resilient, physics-based navigation and timing.